

Image Analysis Report

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Executive Summary

High severity localized corrosion centered on a horizontal weld or joint, showing advanced coating failure and active oxidation. The presence of exfoliating scale suggests significant base metal loss, necessitating immediate integrity verification through non-destructive testing.

Key Findings

#	Finding
1	Severe localized corrosion centered on a horizontal weld bead or structural seam.
2	Presence of thick, exfoliating rust scale indicating advanced material degradation.
3	Extensive coating delamination and blistering spreading from the primary corrosion site.
4	Evidence of crevice corrosion where the horizontal joint has trapped moisture and contaminants.
5	Widespread surface staining suggests active corrosion and lack of cathodic protection or coating integrity.
6	Exact remaining wall thickness and pit depth cannot be determined from the visual image alone.
7	The corrosion pattern indicates a high-risk area for localized thinning and potential through-wall penetration.

Recommendations

#	Recommendation
1	Perform Ultrasonic Thickness (UT) testing across the affected area to determine remaining wall thickness vs. design minimum.
2	Clean the surface to SSPC-SP 10 / NACE No. 2 (Near-White Metal Blast Cleaning) to fully expose the metal profile.
3	Conduct a Pit Depth Gauge survey to quantify the maximum material loss in the localized 'hot spots'.
4	Execute Magnetic Particle Inspection (MPI) on the weld area to identify potential stress corrosion cracking or surface-breaking defects.
5	Apply a multi-coat high-build epoxy coating system compliant with ISO 12944 standards for highly corrosive environments.
6	If wall thinning exceeds the 12.5% allowable limit or code-calculated minimum, perform a structural weld repair or install a corrosion-resistant patch.
7	Verify the integrity of nearby seals or gaskets that may be contributing to moisture entrapment at this joint.